

Shapiro Time Delay derivates from Refraction

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ABSTRACT: The Shapiro effect would be a proof of the theory of Relativity. But it could also be explained by the phenomenon of refraction of gases surrounding the solar corona. According to the theory of Relativity, this effect decreases logarithmically as a function of the distance to the Sun. It is the same with neo-Newtonian mechanics (NNM), except that this decay could be more accentuated and would depend on the frequency of the electromagnetic wave. More precise measurements would need to be done with radar waves and with radio waves.

RESUME : *L'effet Shapiro serait une preuve de la théorie de la Relativité. Mais il pourrait également être expliqué par le phénomène de réfraction des gaz entourant la couronne solaire. Selon la théorie de la Relativité, cet effet décroît de manière logarithmique en fonction de la distance au Soleil. Il en est de même avec la mécanique néo-newtonienne (NNM), si ce n'est que cette décroissance pourrait être plus accentuée et dépendrait de la fréquence de l'onde électro-magnétique. Des mesures plus précises demanderaient à être réalisées avec des ondes radar et avec des ondes radio.*

KEYWORDS: Shapiro effect, time delay, refraction, space-time, against Relativity, alternative neo-Newtonian mechanics.

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1. INTRODUCTION

In the 60s, astronomers sent radar waves (at 7.84 GHz) to Venus to, after crossing the cloud layer, probe its surface. And they realized that the round-trip time of the wave increased as the radio wave passed near the solar corona. This delay time has been nicknamed by Irwin Shapiro

himself as being *the Fourth Test of General Relativity* ^[1].

2. MEASUREMENTS

These measurements carried out ^[2] to demonstrate the Shapiro effect nevertheless call for some remarks:

- a) the measurements are essentially made at low values when the delay is less than 40 μs (see **Figure 1**).
- b) The maximum value (when the photon skims the solar disk) can theoretically go up to 252 μs is an extrapolation, it is not directly measured. The 1970 measurement gives about 192 μs , a difference of 25%
- c) the available graphs are not based on the distance from the Sun but on the days of observation. Then it is not possible to check the Shapiro effect 'directly' as a function of the distance to the Sun.

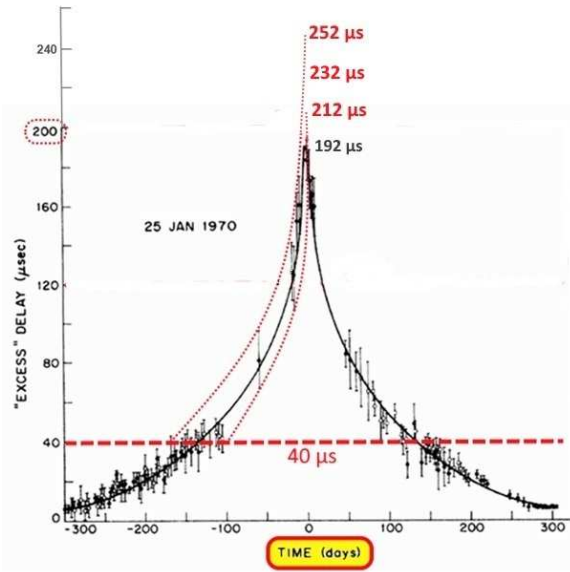


Figure 1: Measurement of Shapiro effect

3. RELATIVISTIC EXPLANATION

According to the theory of relativity, the speed of light being constant, the increase in the journey time comes from an increase in the distance traveled by deformation of the space-time. The relativistic formula of the delay of the Shapiro effect, according to the authors [3] [4] [5] [6] [7] [8], for a round trip near the Sun is/are:

$$\begin{cases} \Delta t(\text{Relativity A}) = 2 \frac{2GM}{c^3} \left[\ln \left(\frac{4R_1 R_2}{R_0^2} \right) + 1 \right] \\ \Delta t(\text{Relativity B}) = 2 \frac{2GM}{c^3} \left[\ln \left(\frac{4R_1 R_2}{R_0^2} - 1 \right) \right] \text{ (1.a-b-c)} \\ \Delta t(\text{Relativity C}) = 2 \frac{2GM}{c^3} \left[\ln \left(\frac{4R_1 R_2}{R_0^2} \right) - 1 \right] \end{cases}$$

with

- R1 the distance from the Sun to the transmitter
- R2 the distance from the Sun to the reflector
- R0 the minimum distance of the photon from the Sun

In all cases, the Shapiro delay is a logarithmic function as a function of the distance from the Sun (it is not a function in $1/R$ for example). On the other hand, the difference of formula depending of the authors is strange and shows the uncertainty of these relativistic functions!

4. NEO-NEWTONIAN MECHANICS EXPLANATION CLOSE TO THE SUN

Neo-Newtonian mechanics (NNM) is Newtonian mechanics without the principle of equivalence; it allows to explain many so-called relativistic phenomena at very high speed (**more information on 9.Related Articles**).

According to neo-Newtonian mechanics, the increase in travel time does not come from an increase in the distance but from a slowing of the speed of light. This slowing down of the speed of light is a classic effect of refraction, refraction that would be caused by the gases surrounding the Sun. This hypothesis is also mentioned by Pr. Taillet at the end of his broadcasted course [3].

The neo-Newtonian formula of the delay of the Shapiro effect is for a round trip close to the Sun (see **Appendix B**):

$$\Delta t(\text{NNM}) = 2 \frac{R_S(n_S-1)E(R)}{c} \left[\ln \left(\frac{4R_1 R_2}{R_0^2} \right) \right] \quad (2)$$

With $E(R)$ decreasing as a function of distance, for example exponentially as can be seen with the gas surrounding the Earth [9].

We find that $\Delta t(\text{NNM})$ is also a logarithmic function as a function of the distance to the Sun. To correlate with relativistic predictions (eg Eq. (1.a)), this would give a refractive index of the order of:

$$n_0 - 1 = \frac{1}{E(R)} \frac{2GM}{R_0 c^2} \left[1 + \frac{1}{\ln \left(\frac{4R_1 R_2}{R_0^2} \right)} \right] \quad (3)$$

The values are then virtually identical to the relativistic predictions (see **Figure 2**).

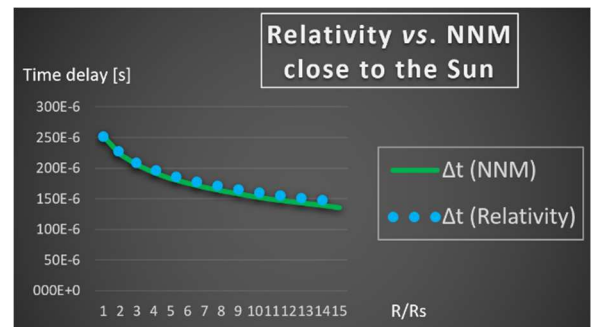


Figure 2: Relativity vs. Neo-Newtonian Mechanics

Nota : Idem for the Eq. (1-b) or (1-c)

5. NNM EXPLANATION FAR FROM THE SUN

We saw in the first part that the measurements were essentially made at values less than 40 μs , when the photon passes beyond 100 solar radii, or practically the distance from the Sun to Venus. (155 solar radii)! It is then necessary to use non-simplified formulas (see Appendix A2 & B2).

$$\Delta t(\text{NNM}) = \frac{2}{c} \frac{R_S(n_S-1)E(R_0)}{\ln\left(\frac{R_0^2}{(R_1-\sqrt{R_1^2-R_0^2})(R_2-\sqrt{R_2^2-R_0^2})}\right)} \quad (4)$$

But the neo-Newtonian equation is based on the value of the refractive index, which is only estimated but which is not measured. Therefore, assuming that $E(R)$ is decreasing:

$$E(R_{\text{close}}) > E(R_{\text{far}}) \quad (5)$$

$$\text{Then } \Delta t(\text{NNM})_{\text{close}} \gg \Delta t(\text{NNM})_{\text{far}} \quad (6)$$

The NNM curve decreases more strongly than the Relativistic curve (for example by 1% at each R_S distance) and can explain that most of the measures are lower than 40 μs (see Figure 3).

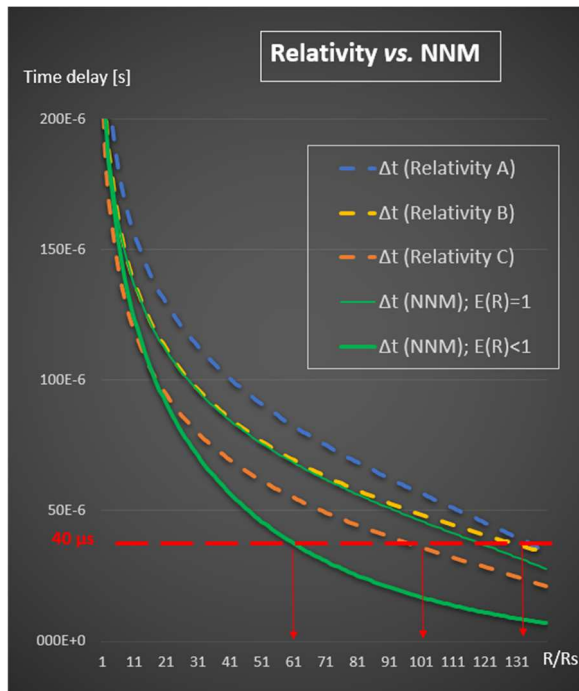


Figure 3: Comparison far from the Sun

6. DISCUSSION ABOUT THE FREQUENCIES

However, according to E. Dowdve, the Shapiro delay is essentially a frequency dependent transit-time effect ^{[10][11]}. And according to Eq. (2) of G. Fontaine ^[12], the transformation of the proper time of D/f^2 , the dispersive delay for propagation

at frequency f along the line of sight. Thus, the delay would also function of the frequency. The concern is that the deformation of the space-time is independent of the wavelength. If the Shapiro delay measurements were confirmed to depend on the frequency f , this could invalidate the notion of space-time...

Moreover, the neo-Newtonian formula does allow a variation of the results according to the frequency. This is why it would be necessary to carry out these observations again with radar waves and with radio waves in order to discriminate between these two theories.

7. CONCLUSION

The Shapiro delay decreases logarithmically as a function of the distance to the Sun. But the only measurements accessible to the public are those made according to the observation days, which does not make it possible to verify with precision the Shapiro delay as a function of a distance to the Sun. And if the relativistic explanation by deformation of the space-time is generally accepted, it can be criticized for its various equations and it is criticized by some for its possible variation according to the wavelength.

We can then advance another explanation: the solar corona being necessarily surrounded by gas, this slight presence of gas would be enough to generate a phenomenon of refraction. The decay curve would be logarithmic as well. Far from the Sun, an exponential decay of the concentration (as for the Earth's atmosphere) would explain that most of the measurements are less than 40 μs . The best way to make sure of this possible refraction phenomenon is to measure the Shapiro delay as a function of the distance to the Sun and especially as a function of frequency, i.e. with radar waves and with radio waves.

8. REFERENCES

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9. RELATED ARTICLES / NEO-NEWTONIAN MECHANICS

EXPERIMENTS:	
How to Demonstrate the Lorentz Factor: Variable Time vs. Variable Inertial Mass	http://file.scirp.org/pdf/JMP_2015022510573131.pdf
Velocity Addition Demonstrated from the Conservation of Linear Momenta, an Alternative Expression	http://file.scirp.org/pdf/JMP_2015050609513342.pdf
Net Force $F = \gamma^3 m a$ at High Velocity	http://file.scirp.org/pdf/JMP_2016042814580505.pdf
An improvement of the accuracy of Fizeau's experiment	http://gsjournal.net/Science-Journals/Research%20Papers-Relativity%20Theory/Download/7247
OBSERVATIONS:	
About the ovoid orbits in general, and perihelion precession of Mercury in particular (2)	http://www.mrelativity.net/Papers/51/Mercury%20Millennium%20Serret%205%20janvier%202018.pdf
Hipparcos did not measure directly the light bending!	http://gsjournal.net/Science-Journals/Research%20Papers-Mechanics%20/%20Electrodynamics/Download/6998
The flat rotation curve of our galaxy explained within Newtonian mechanics	https://physicsessays.org/browse-journal-2/product/1240-7-olivier-serret-the-flat-rotation-curve-of-our-galaxy-explained-within-newtonian-mechanics.html
Gravity vs. Dark Energy, about the Expansion of the Universe	http://file.scirp.org/pdf/JMP_2018011714405269.pdf
Gravitational waves or particle radiation?	https://www.physicsessays.org/browse-journal-2/product/1588-12-olivier-serret-gravitational-waves-or-particle-radiation.html
The Pioneer Anomaly explained by the Processing of the Doppler Effect	http://gsjournal.net/Science-Journals/Research%20Papers-Relativity%20Theory/Download/7330
<i>Shapiro time delay derivates from refraction</i>	<i>Present one</i>
<i>The gravitational light bending: is Relativity unvalid?</i>	<i>Next one</i>
CRITICISM OF RELATIVITY:	
Reply to "A Simple Derivation of the Lorentz Transformation"	http://file.scirp.org/pdf/JMP_2017121915203275.pdf

10. APPENDIXES

A. ACCORDING TO THE THEORY OF THE RELATIVITY

A1. Chart and graph

$X=R1.R2/Rs^2$	Unit	Venus	Mars
R1 : distance Earth-Sun	m	1,47E+11	1,47E+11
R2 : distance Sun-Planet	m	1,08E+11	2,28E+11
Rs : Sun radius	m	7,00E+08	7,00E+08
t : round trip duration	s	1 700	2 498
$\Delta t/a$: delay in $\ln(X)+1$	s	252E-6	266E-6
$\Delta t/b$: delay in $\ln(X-1)$	s	232E-6	247E-6
$\Delta t/c$: delay in $\ln(X)-1$	s	212E-6	227E-6

Chart A: Various theoretical Shapiro time delays

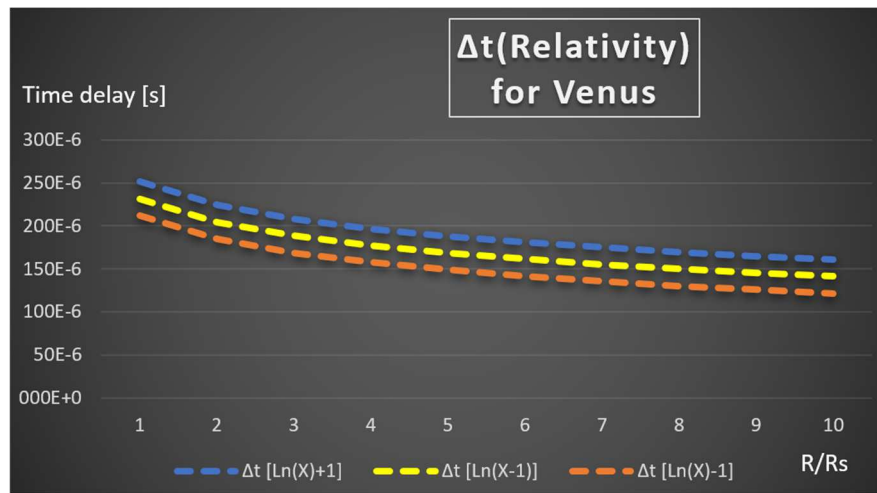


Figure A: Various relativistic formula graphes

A2. Relativistic derivation far from the Sun (or in general)

Let us start from Relativistic Shapiro demonstration [3], with:

$$\text{Schwarzschild radius } a = \frac{2GM}{c^2} \quad (A1)$$

$$T_{1-0} = \frac{1}{c} \left[\sqrt{r^2 - R_0^2} \right]_{R_0}^{R_1} + \frac{a}{c} \left[\ln \left(\frac{r + \sqrt{r^2 - R_0^2}}{R_0} \right) \right]_{R_0}^{R_1} + \frac{a}{2c} \left[\sqrt{\frac{r-R_0}{r+R_0}} \right]_{R_0}^{R_1} \quad (A2)$$

$$T_{1-0} = \frac{1}{c} \sqrt{R_1^2 - R_0^2} + \frac{a}{c} \ln \left(\frac{R_1 + \sqrt{R_1^2 - R_0^2}}{R_0} \right) + \frac{a}{2c} \sqrt{\frac{R_1 - R_0}{R_1 + R_0}} \quad (A3)$$

 Then for $\Delta t(\text{Rel.}) = \text{Relativistic time delay}$, $E=\text{Earth}$, $V=\text{Venus}$ (or another planet)

$$\Delta t(\text{Rel}) = 2 \frac{a}{c} \left\{ \ln \left(\frac{(R_E + \sqrt{R_E^2 - R_0^2})(R_V + \sqrt{R_V^2 - R_0^2})}{R_0^2} \right) + \frac{1}{2} \sqrt{\frac{R_E - R_0}{R_E + R_0}} + \frac{1}{2} \sqrt{\frac{R_V - R_0}{R_V + R_0}} \right\} \quad (A4)$$

B. NEO-NEWTONIAN DERIVATION

B1. NNM derivation close to the Sun

Hypotheses:

$$t = \frac{l}{c_R} \quad (B1)$$

With t = duration

l = length

c_R = celerity of light in the gas at the distance R of the Sun.

By definition of the index of refraction n :

$$n = \frac{c}{c_R} \quad (B2)$$

$$n - 1 = \frac{n_S - 1}{R/R_S} E(R) \quad (B3)$$

$$R_0 = R \sin(\theta) \quad (B4)$$

$$dl \cos(\varphi) = R d\theta \quad (B5)$$

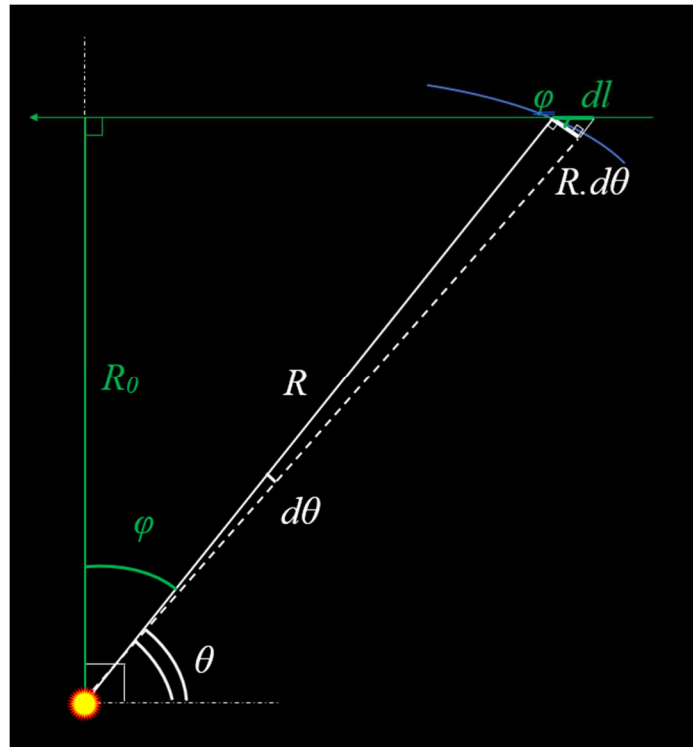


Figure B1

Derivation:

$$\text{From Eq.(B1)} \quad dt = \frac{dl}{c_R} \quad (B6)$$

$$\text{From Eq.(B2)} \quad \frac{1}{c_R} = \frac{n}{c} \quad (B7)$$

$$\text{From Eq.(B3)} \quad n = \frac{R_S(n_S - 1)E(R) + R}{R} \quad (B8)$$

$$\text{From Eq.(B5)} \quad dl = \frac{R}{\cos(\varphi)} d\theta \quad (\text{B9})$$

$$dl = \frac{R}{\cos(\frac{\pi}{2}-\theta)} d\theta \quad (\text{B10})$$

$$dl = \frac{R}{\sin(\theta)} d\theta \quad (\text{B11})$$

$$\text{From Eq.(B6) and (B11)} \quad dt = \frac{1}{c_R} \frac{R}{\sin(\theta)} d\theta \quad (\text{B12})$$

$$\text{With Eq.(B7) and (B8)} \quad dt = \frac{1}{c} \frac{R_S(n_S-1)E(R)+R}{R} \frac{R}{\sin(\theta)} d\theta \quad (\text{B13})$$

$$dt = \left(\frac{R_S(n_S-1)E(R)}{c \sin(\theta)} + \frac{R}{c \sin(\theta)} \right) d\theta \quad (\text{B14})$$

$$\text{With Eq.(B4)} \quad dt = \left(\frac{R_S(n_S-1)E(R)}{c \sin(\theta)} + \frac{R_0}{c \sin^2(\theta)} \right) d\theta \quad (\text{B15})$$

Let us T_{1-0} the duration from the body [1] to the proximity of the Sun [0]:

$$T_{1-0} = \int_{\theta_1}^{\frac{\pi}{2}} \frac{R_0}{c \sin^2(\theta)} d\theta + \int_{\theta_1}^{\frac{\pi}{2}} \frac{R_S(n_S-1)E(R)}{c \sin(\theta)} d\theta \quad (\text{B16})$$

$$T_{1-0} = \frac{R_0}{c} \int_{\theta_1}^{\frac{\pi}{2}} \frac{d\theta}{\sin^2(\theta)} + \frac{R_S(n_S-1)}{c} \int_{\theta_1}^{\frac{\pi}{2}} \frac{E(R)d\theta}{\sin(\theta)} \quad (\text{B17})$$

$$T_{1-0} = \frac{R_0}{c} [-\cot(\theta)]_{\theta_1}^{\frac{\pi}{2}} + \frac{R_S(n_S-1)E(R_0)}{c} \left[\ln\left(\tan\left(\frac{\theta}{2}\right)\right) \right]_{\theta_1}^{\frac{\pi}{2}} \quad (\text{B18})$$

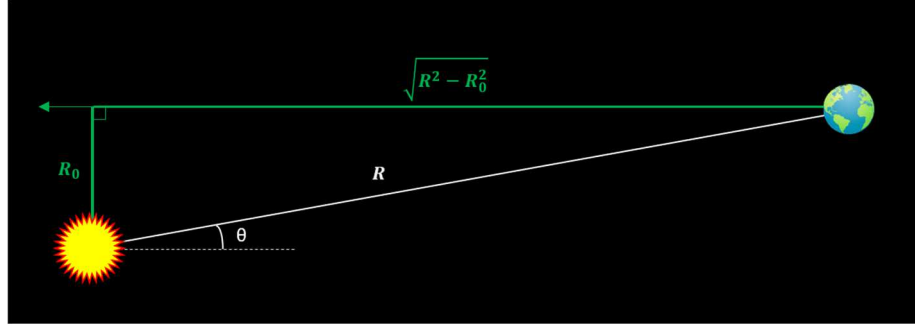


Figure B2

$$\tan(\theta) = \frac{R_0}{\sqrt{R^2 - R_0^2}} \quad (\text{B19})$$

$$\text{For } R_0 \ll R \quad \tan(\theta) \approx \frac{R_0}{R} \quad (\text{B20})$$

$$\text{then} \quad \cot(\theta_1) = \frac{R_1}{R_0} \quad (\text{B21})$$

$$\text{so} \quad [-\cot(\theta)]_{\theta_1}^{\frac{\pi}{2}} = -\left(0 - \frac{R_1}{R_0}\right) = \frac{R_1}{R_0} \quad (\text{B22})$$

$$\left[\ln\left(\tan\left(\frac{\theta}{2}\right)\right) \right]_{\theta_1}^{\frac{\pi}{2}} = \ln\left(\tan\left(\frac{\pi}{4}\right)\right) - \ln\left(\tan\left(\frac{\theta_1}{2}\right)\right) \quad (\text{B23})$$

$$\text{For } R_0 \ll R \quad \left[\ln\left(\tan\left(\frac{\theta}{2}\right)\right) \right]_{\theta_1}^{\frac{\pi}{2}} \approx \ln(1) - \ln\left(\frac{R_0}{2R_1}\right) \quad (\text{B24})$$

$$\ln(1) = 0 \quad \text{then} \quad \left[\ln\left(\tan\left(\frac{\theta}{2}\right)\right) \right]_{\theta_1}^{\frac{\pi}{2}} = \ln\left(\frac{2.R_1}{R_0}\right) \quad (\text{B25})$$

$$\text{From Eq.(B18), (B22) and (B25)} \quad T_{1-0} = \frac{R_0}{c} \frac{R_1}{R_0} + \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\tan\left(\frac{2.R_1}{R_0}\right)\right) \quad (\text{B26})$$

$$T_{1-0} = \frac{R_1}{c} + \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{2.R_1}{R_0}\right) \quad (\text{B27})$$

$$\text{So for the body [2]} \quad T_{2-0} = \frac{R_2}{c} + \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{2.R_2}{R_0}\right) \quad (\text{B28})$$

$$\text{And for a round-trip} \quad T = 2 * (T_{1-0} + T_{2-0}) \quad (\text{B29})$$

$$T = 2 \frac{R_1+R_2}{c} + 2 \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{2.R_1}{R_0} \frac{2.R_2}{R_0}\right) \quad (\text{B30})$$

$$T = 2 \frac{R_1+R_2}{c} + 2 \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{4.R_1.R_2}{R_0^2}\right) \quad (\text{B31})$$

$$\left\{ \begin{array}{l} 2 \frac{R_1+R_2}{c} = \text{normal round trip duration} \\ 2 \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{4.R_1.R_2}{R_0^2}\right) = \Delta t(\text{NNM}) = \text{time delay} \end{array} \right. \quad (\text{B32.a-b})$$

Equivalence

$$\left\{ \begin{array}{l} \text{Relativity:} \quad \Delta t = 2 \frac{2GM}{c^3} \left[\ln\left(\frac{4R_1R_2}{R_0^2}\right) + 1 \right] \\ \text{Neo - Newtonian:} \quad \Delta t = 2 \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{4.R_1.R_2}{R_0^2}\right) \end{array} \right. \quad (\text{B33.a-b})$$

$$2 \frac{2GM}{c^3} \left[\ln\left(\frac{4R_1R_2}{R_0^2}\right) + 1 \right] = 2 \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{4.R_1.R_2}{R_0^2}\right) \quad (\text{B34})$$

$$(n_0 - 1)E(R_0) = \frac{2GM}{R_S.c^2} \left[1 + \frac{1}{\ln\left(\frac{4R_1R_2}{R_0^2}\right)} \right] \quad (\text{B35})$$

B2. Derivation far from the Sun (or in general)

$$\text{In general} \quad \tan\left(\frac{\theta}{2}\right) = \frac{1-\cos(\theta)}{\sin(\theta)} \quad (\text{B36})$$

$$\text{then} \quad \tan\left(\frac{\theta}{2}\right) = \frac{1-\sqrt{R^2-R_0^2}/R}{R_0/R} \quad (\text{B37})$$

$$\tan\left(\frac{\theta}{2}\right) = \frac{R-\sqrt{R^2-R_0^2}}{R_0} \quad (\text{B38})$$

$$\left[\ln\left(\tan\left(\frac{\theta}{2}\right)\right) \right]_{\theta_1}^{\frac{\pi}{2}} = -\ln\left(\tan\left(\frac{\theta_1}{2}\right)\right) = +\ln\left(\frac{R_0}{R_1-\sqrt{R_1^2-R_0^2}}\right) \quad (\text{B39})$$

$$\tan(\theta) = \frac{R_0}{\sqrt{R^2-R_0^2}} \quad (\text{B40})$$

$$\text{so} \quad [-\cot(\theta)]_{\theta_1}^{\frac{\pi}{2}} = -\left(0 - \frac{\sqrt{R_1^2 - R_0^2}}{R_0}\right) = \frac{\sqrt{R_1^2 - R_0^2}}{R_0} \quad (\text{B41})$$

$$\text{From Eq.(B34), (B38) and (B41)} \quad T_{1-0} = \frac{R_0 \sqrt{R_1^2 - R_0^2}}{c} + \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{R_0}{R_1 - \sqrt{R_1^2 - R_0^2}}\right) \quad (\text{B42})$$

$$T_{1-0} = \frac{\sqrt{R_1^2 - R_0^2}}{c} + \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{R_0}{R_1 - \sqrt{R_1^2 - R_0^2}}\right) \quad (\text{B43})$$

$$\text{So for the body [2]} \quad T_{2-0} = \frac{\sqrt{R_2^2 - R_0^2}}{c} + \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{R_0}{R_2 - \sqrt{R_2^2 - R_0^2}}\right) \quad (\text{B44})$$

$$\text{And for a round-trip} \quad T = 2 * (T_{1-0} + T_{2-0}) \quad (\text{B45})$$

$$T = 2 \frac{\sqrt{R_1^2 - R_0^2} + \sqrt{R_2^2 - R_0^2}}{c} + 2 \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{R_0^2}{(R_1 - \sqrt{R_1^2 - R_0^2})(R_2 - \sqrt{R_2^2 - R_0^2})}\right) \quad (\text{B46})$$

$$\left\{ \begin{array}{l} 2 \frac{\sqrt{R_1^2 - R_0^2} + \sqrt{R_2^2 - R_0^2}}{c} = \text{normal round trip duration} \\ 2 \frac{R_S(n_S-1)E(R_0)}{c} \ln\left(\frac{R_0^2}{(R_1 - \sqrt{R_1^2 - R_0^2})(R_2 - \sqrt{R_2^2 - R_0^2})}\right) = \Delta t(\text{NNM}) = \text{time delay} \end{array} \right.$$

(B47.a-b)

Nota: to check this general formula when $R_0 \ll R_1$ and $R_0 \ll R_2$:

$$R_1 - \sqrt{R_1^2 - R_0^2} = R_1 \left\{ 1 - \sqrt{1 - \left(\frac{R_0}{R_1}\right)^2} \right\} \quad (\text{B48})$$

$$R_1 - \sqrt{R_1^2 - R_0^2} \approx R_1 \left\{ 1 - \left(1 - \frac{1}{2} \left(\frac{R_0}{R_1}\right)^2\right) \right\} \quad (\text{B49})$$

$$R_1 - \sqrt{R_1^2 - R_0^2} = R_1 \left\{ \frac{1}{2} \left(\frac{R_0}{R_1}\right)^2 \right\} \quad (\text{B50})$$

$$R_1 - \sqrt{R_1^2 - R_0^2} = \frac{R_0^2}{2.R_1} \quad (\text{B51})$$

$$\ln\left(\frac{R_0^2}{(R_1 - \sqrt{R_1^2 - R_0^2})(R_2 - \sqrt{R_2^2 - R_0^2})}\right) = \ln\left(\frac{R_0^2}{\left(\frac{R_0^2}{2.R_1}\right)\left(\frac{R_0^2}{2.R_2}\right)}\right) \quad (\text{B52})$$

$$\ln\left(\frac{R_0^2}{(R_1 - \sqrt{R_1^2 - R_0^2})(R_2 - \sqrt{R_2^2 - R_0^2})}\right) = \ln\left(\frac{4.R_1.R_2}{R_0^2}\right) \quad (\text{B53})$$

Then we get the same formula than for the previous one close to the Sun (Eq.(B32-b))